DS605: Fundamentals of Machine Learning

Challenge Assignment

Black-Box Hyperparameter Optimization

Credentials: Team ID, API Key, and API URL will be shared on your respective DAU email ID.

Submission: Public GitHub repository link or Jupyter Notebook (.ipynb) file.

Objective

Find the best hyperparameter configuration from the provided search space with the lowest possible loss, using as few model (Oracle) calls as possible.

Rules

- This is an individual challenge.
- Use only the Team ID, API Key, and API URL assigned to you.
- Write your own logic for selecting the next hyperparameter configuration. Do not use automated hyperparameter-optimization/search libraries.
- Every valid Oracle call is counted; repeated valid configurations are also counted.
- Use only the parameter names and values returned by your assigned search space. Lower loss is better.

Task 1 - Check Your Assigned Search Space

Insert the credentials received by email and run this once to see your assigned model and allowed hyperparameter values.

```
import json
from urllib.request import Request, urlopen

TEAM_ID = "YOUR_TEAM_ID"
API_KEY = "YOUR_API_KEY"
API_URL = "YOUR_API_URL"

def api(payload):
    req = Request(API_URL, data=json.dumps(payload).encode(),
                  headers={"Content-Type":"application/json", "X-API-Key":API_KEY})
    with urlopen(req, timeout=30) as r:
        return json.load(r)

spec = api({"action":"spec", "team_id":TEAM_ID})
SPACE = spec["hyperparameters"]
print("Model:", spec["model"])
for name, values in SPACE.items():
    print(name, ":", values)
```

Task 2 - Query the Oracle

Use the same format shown below: choose one allowed value for every parameter in your search space, pass the dictionary to oracle_query(), and it will return the loss.

```
def oracle_query(params):
    result = api({"action":"query", "team_id":TEAM_ID, "params":params})
    return float(result["loss"])

# Example format only - use YOUR Oracle's parameter names and allowed values
params = {
    "criterion": "squared_error",
    "splitter": "best",
    "max_depth": 4,
    "min_samples_split": 5,
    "min_samples_leaf": 2,
    "max_features": 0.8,
}

loss = oracle_query(params)
print("Loss:", loss)
```

Submission

- Submit your complete runnable notebook/code showing your own optimization logic, queried configurations and losses, final best hyperparameters, best loss, and total number of Oracle calls used.
- Submit either a public GitHub repository link or attach the final .ipynb file, as instructed.
- Remove your API Key from any public GitHub repository before submission.